

Calcium/PTH/VitD — The PTHrP Tumor Imposter



1. Ca sensed -> PTH off

WHAT YOU SEE

Top banner: Ca sensed turns PTH off, with a down-arrow.

WHAT IT ENCODES

In humoral hypercalcemia of malignancy, the parathyroid CaSR works normally: it senses the high calcium driven by tumor PTHrP and correctly SUPPRESSES endogenous PTH. So the diagnostic signature is hypercalcemia with a LOW (appropriately suppressed) PTH. This intact feedback is exactly what separates malignancy-driven hypercalcemia from primary hyperparathyroidism, where PTH is inappropriately high.

BOARD TRAP

A stem reports hypercalcemia and assumes the parathyroids must be overactive — WRONG in HHM. The healthy parathyroids shut PTH off in response to the tumor's calcium load. The discriminator: high Ca + LOW PTH points away from a parathyroid source and toward malignancy (PTHrP), vitamin D, or other non-PTH causes.

2. SCC squamous tumor

WHAT YOU SEE

A green monster labeled SCC / squamous on a 'wanted' street.

WHAT IT ENCODES

Squamous cell carcinoma is the CLASSIC PTHrP secretor — squamous cancers of the LUNG, head and neck, esophagus, and cervix. Squamous cell lung cancer is the prototypical board answer for PTHrP-mediated hypercalcemia (whereas small cell lung cancer is associated with SIADH and ectopic ACTH instead). Mnemonic: 'Squamous Cells Spew PTHrP.'

BOARD TRAP

A lung-cancer stem with hypercalcemia where you reflexively pick small cell carcinoma — WRONG for hypercalcemia. SQUAMOUS cell carcinoma causes PTHrP hypercalcemia; small cell causes SIADH/ectopic ACTH. The discriminator: match the lung histology to the paraneoplastic syndrome — squamous → high calcium (PTHrP).

3. RCC renal

WHAT YOU SEE

A crowned red figure labeled RCC / renal.

WHAT IT ENCODES

Renal cell carcinoma is a common PTHrP-secreting tumor causing humoral hypercalcemia and is a notorious paraneoplastic chameleon (also EPO → polycythemia, renin → hypertension). It belongs to the classic PTHrP group alongside squamous carcinomas. Hypercalcemia in RCC may be humoral (PTHrP) or, with bone metastases, locally osteolytic.

BOARD TRAP

A stem with RCC, hypercalcemia, and an elevated PTHrP that you misread as primary hyperparathyroidism — WRONG. PTH is suppressed and PTHrP is elevated, marking malignancy. The discriminator: a separate PTHrP assay, not the standard PTH assay, identifies the culprit; PTH itself is low.

4. Breast / bladder / ovarian

WHAT YOU SEE

Pink figures labeled Breast Ca, Bladder, Ovarian.

WHAT IT ENCODES

Breast, bladder, and ovarian carcinomas also secrete PTHrP. Breast cancer is special because it can cause hypercalcemia by TWO mechanisms: humoral (circulating PTHrP) and local osteolytic (bone metastases producing PTHrP/cytokines locally to drive focal resorption). Recognizing dual mechanisms matters because the workup and the imaging implications differ.

BOARD TRAP

A breast-cancer stem with hypercalcemia where you assume bone metastases are always the cause — WRONG/incomplete. Breast cancer can cause hypercalcemia via circulating PTHrP even without extensive bone mets. The discriminator: humoral PTHrP can be present with minimal bone disease, so check PTHrP and a bone survey/scan rather than assuming local osteolysis.

5. Fake knight / PTHrP imposter

WHAT YOU SEE

A hooded knight labeled PTH/PTHrP holding a key, called a 'fake knight'.

WHAT IT ENCODES

PTHrP is a molecular impostor: it shares N-terminal homology with PTH and 'keys into' the same PTH1R receptor on bone and kidney, reproducing PTH's calcium-raising actions — yet it is a distinct gene product not detected by standard PTH immunoassays. Physiologically PTHrP also has roles in cartilage/bone development, smooth muscle relaxation, and lactation, but in malignancy it acts as an ectopic PTH mimic.

BOARD TRAP

A stem expects the routine PTH (intact PTH) assay to be elevated in PTHrP-mediated hypercalcemia — WRONG. Standard PTH immunoassays do NOT cross-react with PTHrP, so measured PTH is LOW; you must order a SEPARATE PTHrP assay. The discriminator: low PTH does not exclude a PTH-like hormone — send PTHrP when malignancy is suspected.

6. PTH = PTHrP identity panel

WHAT YOU SEE

Top-right: two knights, PTH and PTHrP, set equal, noting same phosphaturia, opposite endogenous PTH.

WHAT IT ENCODES

PTHrP reproduces PTH's downstream effects through PTH1R: hypercalcemia and renal phosphate wasting (hypophosphatemia), plus increased bone resorption. The KEY difference is the endogenous PTH level — high/inappropriate in primary hyperparathyroidism, but SUPPRESSED in PTHrP-driven malignancy. One point of divergence: PTHrP tends to raise calcitriol less than primary HPT, so 1,25-D is often normal/low in HHM.

BOARD TRAP

A stem uses the matching Ca/PO4 pattern (high Ca, low PO4) to conclude primary hyperparathyroidism in a cancer patient — WRONG. PTHrP gives the same Ca/PO4 pattern but with SUPPRESSED PTH. The discriminator: phosphaturia and hypophosphatemia look identical, so the PTH level (low in HHM vs high in primary HPT) is what separates them.

7. PTH1R receptor post

WHAT YOU SEE

A lamppost labeled PTH1R receptor that PTHrP keys into.

WHAT IT ENCODES

Both PTH and PTHrP signal through the shared PTH1R (a Gs-coupled receptor) on osteoblasts and renal tubular cells, raising cAMP to drive calcium reabsorption, phosphaturia, and (via osteoblast RANKL) osteoclastic resorption. Because the receptor and signaling are identical, the biochemical effects on calcium and phosphate are indistinguishable; only the source hormone and the PTH assay differ.

BOARD TRAP

A stem implies PTHrP works through a DIFFERENT receptor or pathway than PTH, producing a different electrolyte pattern — WRONG. They share PTH1R and produce the same Ca/PO4 effects. The discriminator: identical receptor → identical physiology, so distinguish by assay (PTH vs PTHrP), not by the calcium/phosphate pattern.

8. Bone-vault resorption

WHAT YOU SEE

Figures pulling gold (calcium) out of a bone vault.

WHAT IT ENCODES

PTHrP stimulates osteoblasts (via PTH1R) to upregulate RANKL, activating osteoclasts to resorb bone and release calcium into the blood — the same RANKL-mediated indirect mechanism PTH uses. This osteoclast-driven resorption is why anti-resorptives are central to treating hypercalcemia of malignancy: IV bisphosphonates (zoledronic acid 4 mg) are first-line, with denosumab (anti-RANKL) for bisphosphonate failure or renal impairment, after IV saline rehydration.

BOARD TRAP

A stem treats severe hypercalcemia of malignancy with bisphosphonate alone and expects immediate normalization — WRONG/incomplete. The first step is aggressive IV normal saline volume repletion (and calcitonin for rapid short-term lowering); bisphosphonates take 2-4 days to act. The discriminator: rehydrate first, calcitonin bridges, bisphosphonate/denosumab give durable control.

9. Humoral hypercalcemia label

WHAT YOU SEE

A banner across the lower-right reading HUMORAL HYPERCALCEMIA OF MALIGNANCY, arched over the row of treasure chests — names the PTHrP-driven, circulating-hormone mechanism (the most common cause of cancer hypercalcemia).

WHAT IT ENCODES

Humoral hypercalcemia of malignancy (HHM) is PTHrP-mediated, circulating-hormone-driven hypercalcemia and is the MOST COMMON mechanism of malignancy-associated hypercalcemia (~80%), typically with solid tumors (squamous, renal, breast). It generally signals advanced disease and a poor prognosis. It is distinguished mechanistically from local osteolytic hypercalcemia and from 1,25-D-mediated hypercalcemia.

BOARD TRAP

A stem lumps all cancer hypercalcemia into 'bone metastases' — WRONG. Three mechanisms exist: (1) HUMORAL (PTHrP, solid tumors, suppressed PTH, normal/low 1,25-D), (2) LOCAL OSTEOLYTIC (bone mets / multiple myeloma releasing local cytokines), and (3) 1,25-D-MEDIATED (lymphoma, granulomas making calcitriol → high 1,25-D). The discriminator: check PTHrP and 1,25-D to assign the mechanism; HHM is humoral, not metastatic.

10. Low Ca-serum chest (up Ca)

WHAT YOU SEE

A treasure chest spilling gold coins, marked '↑Ca serum' — the overflowing treasure = high serum calcium, the defining lab of PTHrP secretion.

WHAT IT ENCODES

Hypercalcemia is the defining lab abnormality of PTHrP secretion and produces the classic 'stones, bones, groans, psychiatric overtones' (nephrolithiasis, bone pain, abdominal pain/constipation, confusion), plus polyuria and shortened QT on ECG. Severity guides urgency: total calcium >14 mg/dL or symptomatic hypercalcemia is an emergency requiring aggressive IV saline and anti-resorptive therapy.

BOARD TRAP

A stem dismisses moderate hypercalcemia as needing no acute treatment regardless of symptoms — WRONG. Symptomatic hypercalcemia or calcium >14 mg/dL warrants emergent IV saline (and calcitonin/bisphosphonate). The discriminator: treat based on calcium level AND symptoms, not the absolute number alone.

11. Low PO4 serum chest

WHAT YOU SEE

A treasure chest marked '↓PO4 serum' — depleted/half-empty chest representing the renal phosphate wasting (hypophosphatemia) that PTHrP causes via PTH1R.

WHAT IT ENCODES

Like PTH, PTHrP causes renal phosphate wasting via PTH1R inhibition of the proximal Na-Pi cotransporter, producing hypophosphatemia. This shared phosphaturia is why the Ca/PO4 pattern of HHM (high Ca, low PO4) is identical to primary hyperparathyroidism and helps separate it from osteolytic or vitamin-D-mediated hypercalcemia, where phosphate is often normal or high.

BOARD TRAP

A stem uses normal/high phosphate to support a PTHrP diagnosis — WRONG. PTHrP causes LOW phosphate (renal wasting), just like PTH. The discriminator: low phosphate fits PTH/PTHrP-driven hypercalcemia, whereas vitamin-D-mediated and many osteolytic causes do not waste phosphate the same way.

12. Low endogenous PTH chest

WHAT YOU SEE

A treasure chest marked '↓endogenous PTH' beside an '↑PTHrP' tag — native PTH chest emptied (suppressed) while the imposter PTHrP rises.

WHAT IT ENCODES

The complete diagnostic pattern of PTHrP-mediated HHM is: HIGH calcium, LOW phosphate, SUPPRESSED endogenous PTH, and ELEVATED PTHrP (with normal/low 1,25-D). This four-part fingerprint is the high-yield board pattern and immediately distinguishes malignancy from primary hyperparathyroidism, where PTH is the one value that is high instead of suppressed.

BOARD TRAP

A stem gives high calcium and asks the next test, and you stop at a normal PTH thinking the workup is done — WRONG. With high Ca and LOW/suppressed PTH, the next step is PTHrP and 1,25-D levels (and consider malignancy workup). The discriminator: high Ca + high PTH = primary HPT; high Ca + low PTH = check PTHrP (malignancy), vitamin D, and other non-PTH causes.